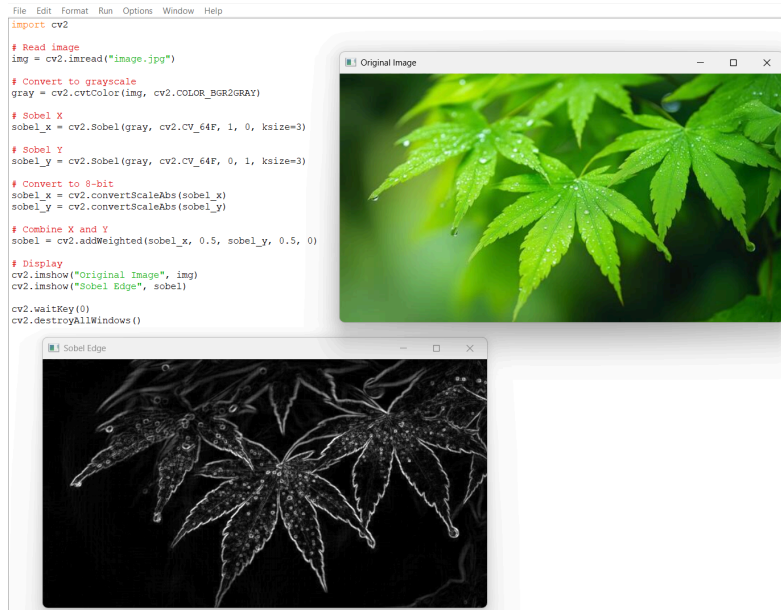


LAB EXPERIMENTS

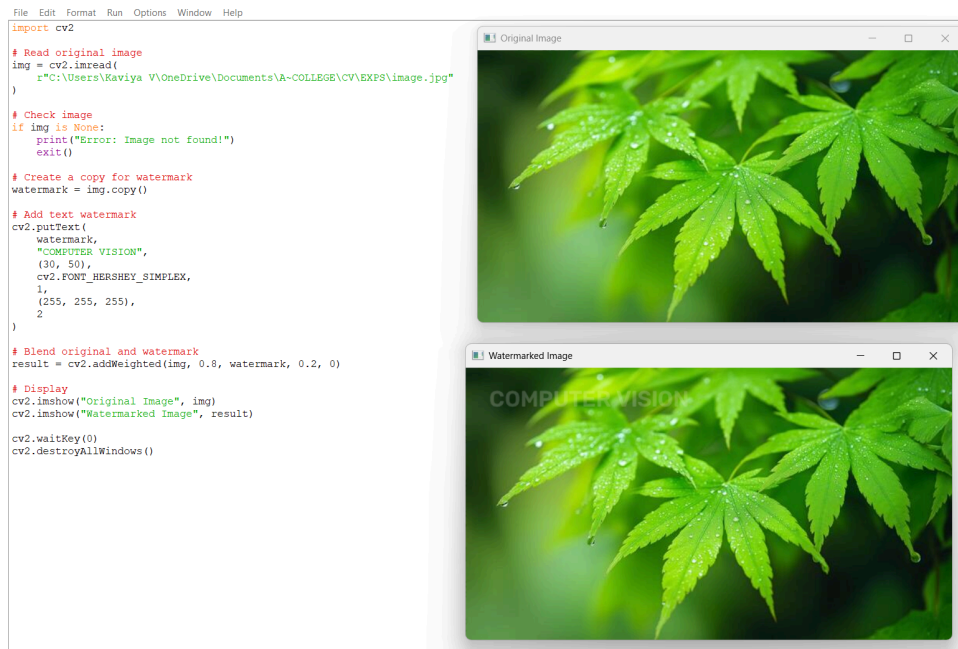
NAME: KAVIYA V
COURSE CODE: ITA0512

REG NO: 192421022
COURSE NAME: COMPUTER VISION

16. Implement a Sobel algorithm using Open CV to filter the input image.



17. Design and implement a water marking technique to insert the watermark into the originalL effectively image using Open CV.



18. Implement image cropping, copying and pasting to select a region of interest (ROI) from the source image using Open CV.

```
import cv2

# Read image
img = cv2.imread(
    r"C:\Users\Kaviya V\OneDrive\Documents\A-COLLEGE\CV\EXPS\image.jpg"
)

# Check image
if img is None:
    print("Error: Image not found!")
    exit()

# Crop ROI
roi = img[50:200, 50:200]

# Copy ROI
copy = roi.copy()

# Get image dimensions
h, w = img.shape[:2]

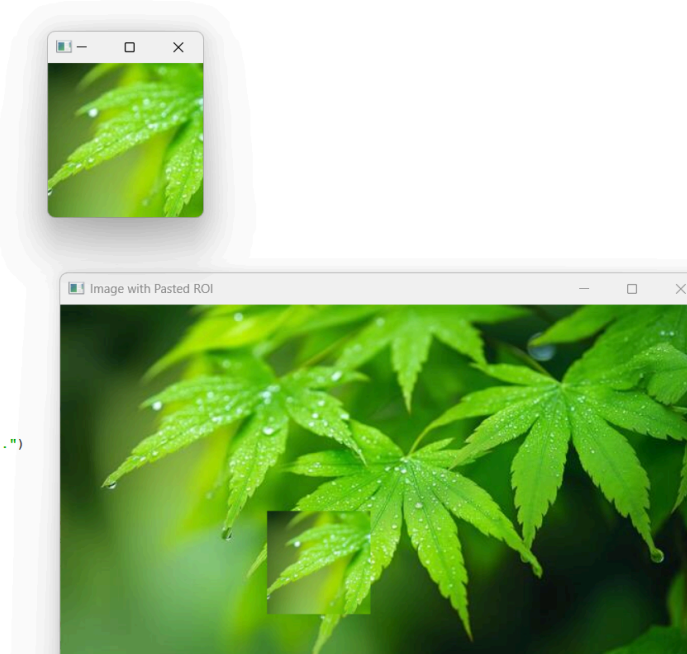
# Resize ROI to a smaller size
copy = cv2.resize(copy, (100, 100))

# Paste at a safe position
x = 200
y = 200

# Check if paste area fits
if y + 100 <= h and x + 100 <= w:
    img[y:y+100, x:x+100] = copy
else:
    print("Paste area is outside the image.")
    exit()

# Display
cv2.imshow("Cropped ROI", roi)
cv2.imshow("Image with Pasted ROI", img)

cv2.waitKey(0)
cv2.destroyAllWindows()
```



19. Implement the Erosion Morphological operations technique using Open CV in python.

```
import cv2
import numpy as np

# Read image
img = cv2.imread(
    r"C:\Users\Kaviya V\OneDrive\Documents\A-COLLEGE\CV\EXPS\image.jpg"
)

# Check image
if img is None:
    print("Error: Image not found!")
    exit()

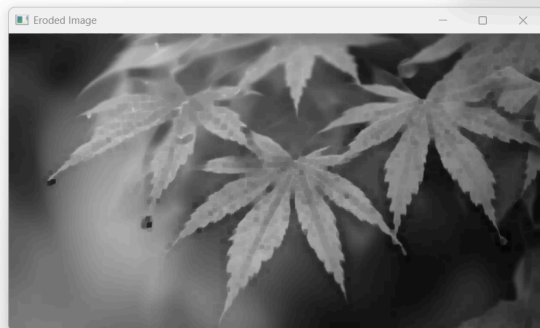
# Convert to grayscale
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

# Create kernel
kernel = np.ones((5, 5), np.uint8)

# Apply erosion
eroded = cv2.erode(gray, kernel, iterations=1)

# Display images
cv2.imshow("Original Image", gray)
cv2.imshow("Eroded Image", eroded)

cv2.waitKey(0)
cv2.destroyAllWindows()
```



20. Implement the Dilation technique as a Morphological operation to dilate the foreground regions based on Open CV.

```
import cv2
import numpy as np

# Read image
img = cv2.imread(
    r"C:\Users\Kaviya V\OneDrive\Documents\A~COLLEGE\CV\EXPS\image.jpg"
)

# Check image
if img is None:
    print("Error: Image not found!")
    exit()

# Convert to grayscale
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

# Create kernel
kernel = np.ones((5, 5), np.uint8)

# Apply dilation
dilated = cv2.dilate(gray, kernel, iterations=1)

# Display images
cv2.imshow("Original Image", gray)
cv2.imshow("Dilated Image", dilated)

cv2.waitKey(0)
cv2.destroyAllWindows()
```

